SWIFTLOG LOGISTICS

Supply Chain Analytics & Business Intelligence

Data Analytics Track | October 2026

PART 1 — Data Import, Understanding & Preparation

Purpose: establish the source-data inventory, inspect the datasets, validate keys and relationships, and define the analytical model.

1.1 Datasets imported

Dataset	Rows	Main purpose
Customers	1,000	Customer demographics, location and purchasing behaviour.
Orders	5,000	Order trends, status and customer purchasing.
Order Items	15,000	Product quantities and order-line revenue.
Products	250	Product category and unit pricing.
Suppliers	150	Supplier master data.
Warehouses	15	Warehouse locations.
Inventory	4,000	Warehouse-product stock records.
Shipments	5,000	Shipment and warehouse activity.
Delivery Performance	5,000	Delivery status and delay days.
Employees	200	Workforce allocation by warehouse.
Returns	500	Returned orders and return reasons.

1.2 Primary and foreign keys

- Orders: OrderID is unique.
- Order Items: OrderItemID is unique; OrderID and ProductID are foreign keys.
- Customers: CustomerID is unique.
- Products: ProductID is unique.
- Shipments: ShipmentID is unique; OrderID and WarehouseID are foreign keys.
- Delivery Performance: ShipmentID is unique and links to Shipments.
- Returns: ReturnID is unique; OrderID links to Orders.
- Inventory: WarehouseID and ProductID identify the warehouse-product combination, but the combination is not unique.
- Warehouses: WarehouseID is unique.
- Employees: EmployeeID is unique; WarehouseID links to Warehouses.
- Suppliers: SupplierID is unique, but it is not linked to an operational transaction table in the supplied data.

1.3 Relationship validation

Relationship	Coverage
Orders.CustomerID → Customers.CustomerID	100%
Order_Items.OrderID → Orders.OrderID	100%
Order_Items.ProductID → Products.ProductID	100%
Shipments.OrderID → Orders.OrderID	100%
Delivery_Performance.ShipmentID → Shipments.ShipmentID	100%
Inventory.ProductID → Products.ProductID	100%
Inventory.WarehouseID → Warehouses.WarehouseID	100%
Employees.WarehouseID → Warehouses.WarehouseID	100%
Returns.OrderID → Orders.OrderID	100%

1.4 Proposed analytical model

- Customer dimension → Orders fact.
- Date dimension → Orders fact.
- Order dimension → Order Items and Shipments.
- Product dimension → Order Items and Inventory.
- Warehouse dimension → Shipments, Inventory and Employees.
- Shipment dimension → Delivery Performance.
- Returns → Orders.

Power BI Relationship Model

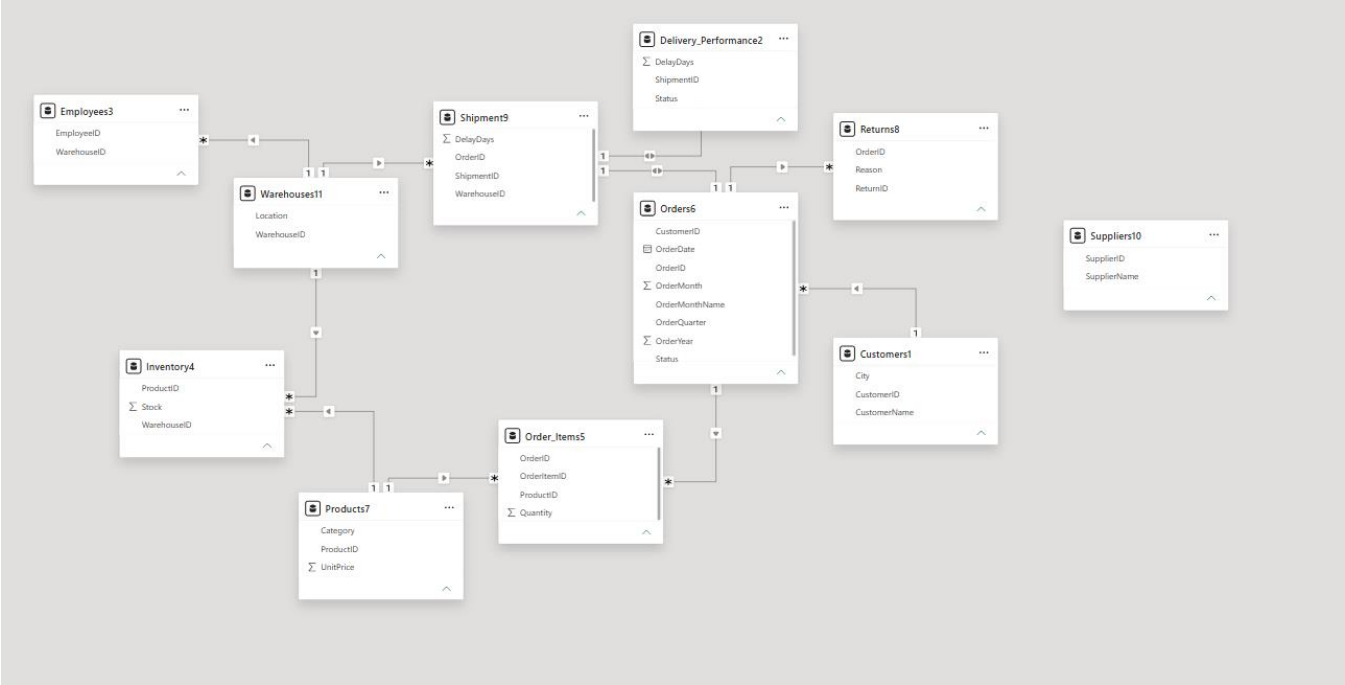


Figure 1. Proposed Power BI data model.

1.5 Data dictionary

The source data dictionary was used to understand supplied fields. Additional analytical fields introduced in the prepared Orders table are documented below.

Field	Definition
OrderYear	Year extracted from OrderDate.
OrderMonth	Numeric month extracted from OrderDate.
OrderMonthName	Month name extracted from OrderDate.
OrderQuarter	Calendar quarter derived from OrderDate.

PART 2 — Data Cleaning & Transformation

Purpose: prepare analysis-ready datasets while preserving valid source information and documenting every transformation.

2.1 Original versus prepared data

Dataset	Original	Prepared	Transformation
Customers	1,000	1,000	Validated; no substantive change.
Orders	5,000	5,000	Added Year, Month, Month Name and Quarter.
Order Items	15,000	15,000	Validated; no substantive change.
Products	250	250	Validated; no substantive change.
Shipments	5,000	5,000	Validated; no substantive change.
Delivery Performance	5,000	5,000	Validated; no substantive change.
Returns	500	500	Validated; no substantive change.
Inventory	4,000	3,997	Removed 3 exact duplicate rows.
Warehouses	15	15	Validated; no substantive change.
Employees	200	200	Validated; no substantive change.
Suppliers	150	150	Validated; no substantive change.

2.2 Cleaning actions

- Inspected every dataset and its columns.
- Validated data types and key fields.
- Checked exact duplicate records.
- Removed 3 exact duplicate Inventory rows.
- Checked foreign-key coverage across the analytical relationships.
- Preserved source values where there was no evidence of an error.
- Created OrderYear, OrderMonth, OrderMonthName and OrderQuarter from OrderDate.
- Prepared analysis-ready CSV outputs.
- Documented unsupported KPIs instead of creating artificial values.

2.3 Inventory grain issue

The cleaned Inventory table still contains repeated WarehouseID + ProductID combinations. There are 2,455 unique combinations and 1,545 repeated occurrences. The source has no inventory date, snapshot ID, lot ID or movement field, so these records cannot safely be treated as historical snapshots or summed without a business rule. The Inventory file contains 3 rows where every field is identical to another row. These are genuine duplicate records and were removed from the cleaned version.

2.4 Source-data gaps

Required analysis	Missing source field	Result
Supplier Performance	SupplierID link to operational transactions	Cannot calculate defensibly.
Warehouse Utilization	Warehouse capacity	Cannot calculate true utilization.
Inventory Turnover	Inventory movement or dated snapshots	Cannot calculate true turnover.
Product Return Rate	ProductID / ReturnItemID in Returns	Cannot attribute returns to products.
Formal retention	Customer signup/cohort date	Repeat-customer rate is possible; cohort retention is not.
Profitability	Product/supplier cost	Revenue is available; margin is not.

SQL

Executive KPIs

```
22      -- Executive KPIs
23      SELECT
24          (SELECT COUNT(DISTINCT OrderID)
25           FROM orders) AS total_orders,
26
27          (SELECT COUNT(DISTINCT CustomerID)
28           FROM customers) AS total_customers,
29
30          (SELECT COUNT(DISTINCT ProductID)
31           FROM products) AS total_products,
32
33          ROUND(
34              (SELECT SUM(oi.Quantity * p.UnitPrice)
35               FROM order_items oi
36               JOIN products p
37                 ON oi.ProductID = p.ProductID), 2
38          ) AS gross_revenue,
39
40          ROUND(
41              (SELECT SUM(oi.Quantity * p.UnitPrice)
42               FROM order_items oi
43               JOIN products p
44                 ON oi.ProductID = p.ProductID)
45              /
46              (SELECT COUNT(DISTINCT OrderID)
47               FROM orders), 2
48          ) AS avg_order_value;
49
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_orders	total_customers	total_products	gross_revenue	avg_order_value
5000	1000	250	21668065.85	4333.61

Monthly Revenue

```sql
-- Monthly revenue and order trend
SELECT
    FORMAT(o.OrderDate, '%Y-%m') AS Transaction_Month,
    COUNT(DISTINCT o.OrderID) AS orders,
ROUND(COALESCE(SUM(oi.Quantity * p.UnitPrice), 0),2) AS revenue
FROM orders o
LEFT JOIN order_items oi ON o.OrderID = oi.OrderID
LEFT JOIN products p ON oi.ProductID = p.ProductID
GROUP BY FORMAT(o.OrderDate, '%Y-%m')
ORDER BY Transaction_Month;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Transaction_Month	orders	revenue
1	594	2582349.75
2	577	2509115.96
3	597	2598402.53
4	638	2742192.82
5	612	2716516.13
6	579	2545277.8
7	595	2535240.76
8	654	2850476.97
9	154	588493.13

Delivery performance by warehouse

```sql
-- Delivery performance by warehouse
SELECT
  w.WarehouseID,
  w.Location,
  COUNT(s.ShipmentID) AS shipments,
  ROUND(AVG(dp.DelayDays),2) AS avg_delay_days,
  ROUND(100.0*SUM(CASE WHEN dp.Status='On Time' THEN 1 ELSE 0 END)/COUNT(*),2) AS on_time_rate_pct
FROM warehouses w
JOIN shipments s ON w.WarehouseID=s.WarehouseID
JOIN delivery_performance dp ON s.ShipmentID=dp.ShipmentID
GROUP BY w.WarehouseID, w.Location
ORDER BY on_time_rate_pct DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

WarehouseID	Location	shipments	avg_delay_days	on_time_rate_pct
W01	Abuja	332	3.46	41.27
W10	Kano	334	3.49	40.12
W07	Lagos	371	3.47	39.62
W14	Lagos	340	3.63	38.53
W04	Lagos	354	3.52	36.72
W11	Lagos	318	3.44	36.48
W15	Kano	352	3.49	36.36
W08	Kano	332	3.67	36.14
W02	Abuja	322	3.44	36.02
W12	Abuja	337	3.59	35.91
W13	Lagos	279	3.61	35.13
W06	Lagos	319	3.61	34.17
W03	Abuja	344	3.68	32.56
W09	Lagos	332	3.68	32.53
W05	Lagos	334	3.79	32.04

Return reasons

```sql
-- Return reasons
SELECT Reason, COUNT(*) AS return_records
FROM returns
GROUP BY Reason
ORDER BY return_records DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Reason	return_records
Late Delivery	185
Wrong Item	163
Damaged	152

PART 3 — KPI Development & DAX Measures

Purpose: create business-focused measures and clearly distinguish supported KPIs from unsupported measures.

KPI	Result	Business value
Total Revenue	NGN 21,668,065.85	Measures gross sales value.
Delivered Revenue	NGN 7,107,470.53	Shows revenue from delivered orders.
Total Orders	5,000	Measures order volume.
Delivered Orders	1,658	Measures completed delivery volume.
Total Customers	1,000	Measures customer base.
Total Products	250	Measures product assortment.
Total Shipments	5,000	Measures logistics volume.
Average Order Value	NGN 4,333.61	Shows average order value.
Delivered AOV	NGN 4,286.77	Shows average delivered order value.
Average Delivery Delay	3.57 days	Measures delivery delay severity.
On-Time Delivery Rate	36.28%	Core delivery reliability KPI.
Return Order Rate	9.46%	Measures returned-order incidence.
Repeat Customer Rate	95.96%	Measures repeat purchasing behaviour.

3.1 Required KPIs that are not defensible from the supplied data

- Inventory Turnover: no inventory movement or dated stock snapshots.
- Supplier Performance Score: no SupplierID connection to operational transactions.
- Warehouse Utilization: no warehouse capacity field.
- Product-level Return Rate: Returns has no ProductID or ReturnItemID.

3.2 DAX measure set

```
SwiftLog Power BI DAX Measures


Total Revenue = SUMX(

    Order_Items, Order_Items[Quantity] * RELATED(Products[UnitPrice])

)

Total Orders =
DISTINCTCOUNT(FactOrders[OrderID])

Total Customers =
DISTINCTCOUNT(DimCustomer[CustomerID])

Total Products =
DISTINCTCOUNT(DimProduct[ProductID])

Total Shipments =
DISTINCTCOUNT(FactShipment[ShipmentID])

Average Order Value =
DIVIDE([Total Revenue], [Total Orders])

Delivered Revenue =
CALCULATE(
    [Total Revenue],
    FactOrders[Status] = "Delivered"
)
```

```
Delivered Orders =
CALCULATE(
    [Total Orders],
    FactOrders[Status] = "Delivered"
)

Delivered AOV =
DIVIDE([Delivered Revenue], [Delivered Orders])



On-Time Delivery Rate =
DIVIDE(
    CALCULATE([Total Shipments], FactDelivery[Status] = "On Time"),
    [Total Shipments]
)

Delayed Shipments =
CALCULATE(
    [Total Shipments],
    FactDelivery[Status] = "Delayed"
)

Return Orders =
DISTINCTCOUNT(FactReturns[OrderID])

Return Order Rate =
DIVIDE([Return Orders], [Total Orders])

Reported Inventory Stock =
SUM(FactInventory[Stock])

Low Stock Records =
CALCULATE(
    COUNTROWS(FactInventory),
    FactInventory[Stock] <= 50
)

Shipments per Employee =
DIVIDE(
    [Total Shipments],
    DISTINCTCOUNT(DimEmployee[EmployeeID])
)

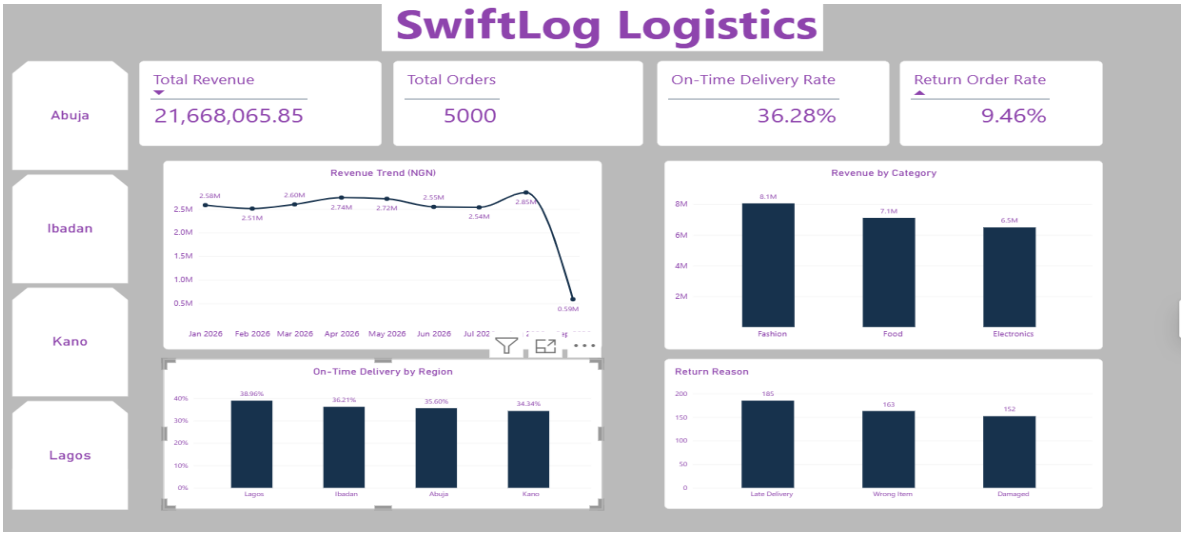
Repeat Customer Rate =
VAR RepeatCustomers =
    COUNTROWS(
        FILTER(
            VALUES(DimCustomer[CustomerID]),
            CALCULATE([Total Orders]) >= 2
        )
    )
RETURN
DIVIDE(RepeatCustomers, [Total Customers])
```

PART 4 — Dashboard Development

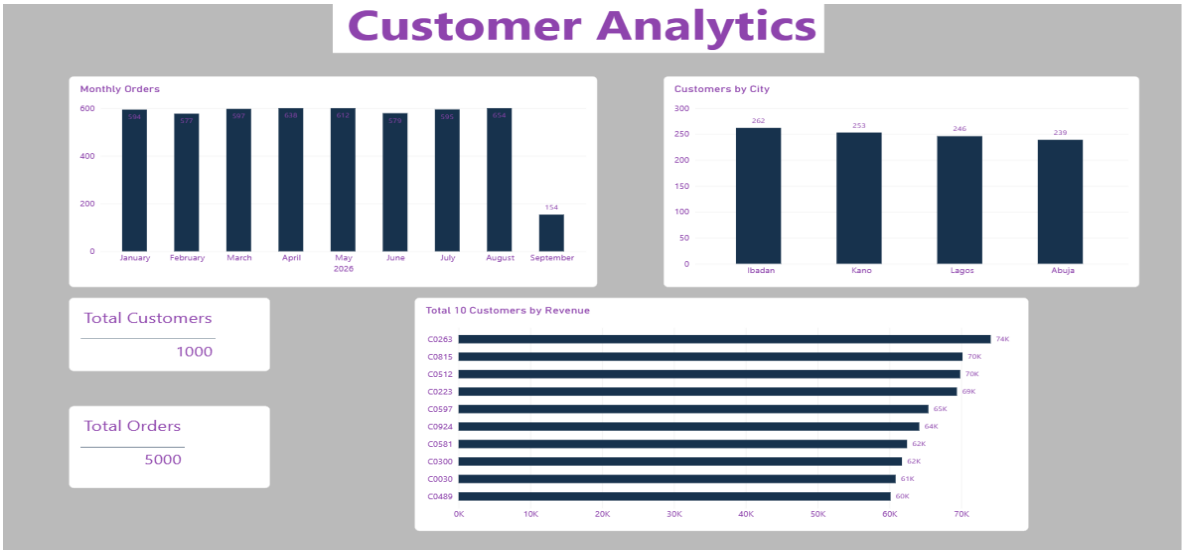
Purpose: deliver five interactive Power BI pages aligned to the assignment structure.

Page	Required focus	Implemented design
Executive Dashboard	KPIs, revenue, orders, delivery, returns, filters	KPI cards, revenue trend, monthly orders, delivery and returns overview.
Customer Analytics	Top customers, revenue, frequency, geography, growth	Customer ranking, revenue, order frequency, city analysis and repeat rate.
Inventory & Warehouse	Stock, capacity, turnover, low stock, comparison	Stock monitoring, low-stock analysis, warehouse comparison and staffing-load proxy.
Shipment & Delivery	Status, delays, region, average delivery, trend	Delivery status, delay analysis, regional and warehouse performance.
Supplier & Product	Supplier, categories, best sellers, returns, revenue	Category/product analysis plus explicit supplier-data limitation.

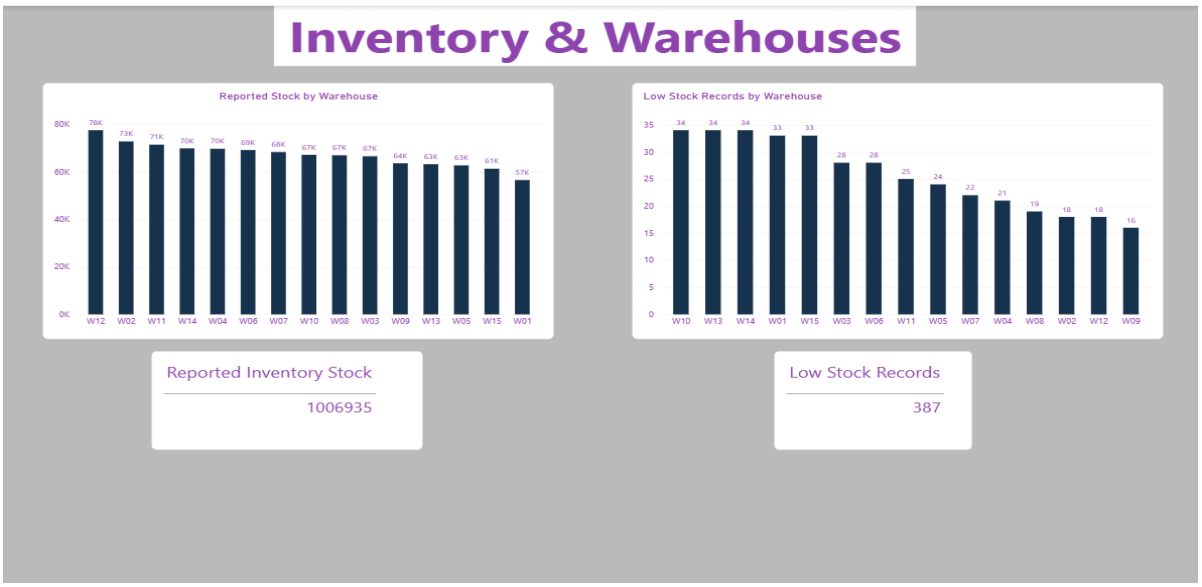
Executive Dashboard



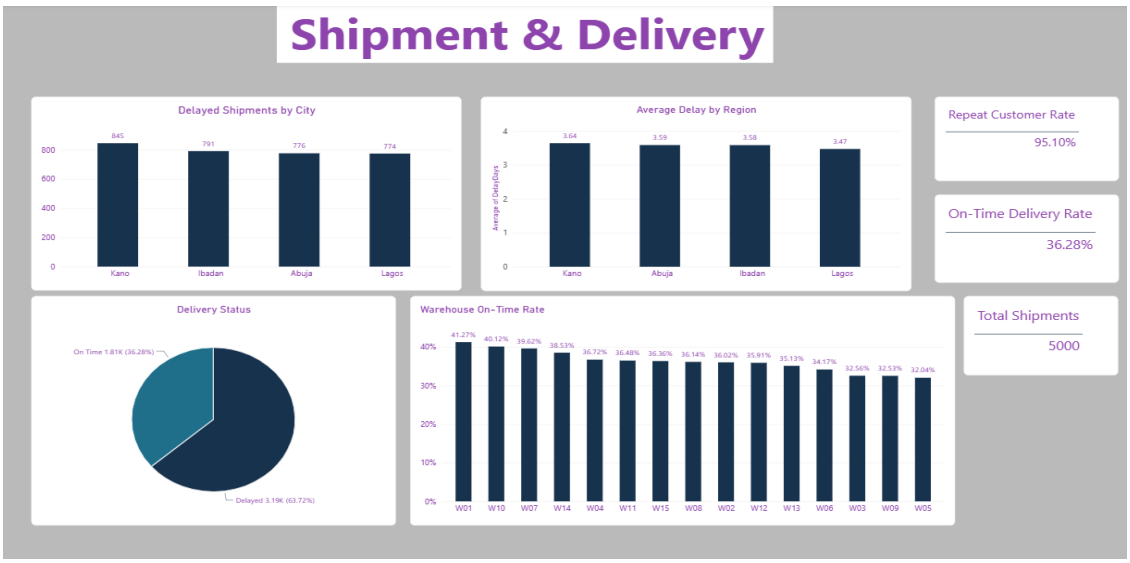
Customer Analytics Dashboard



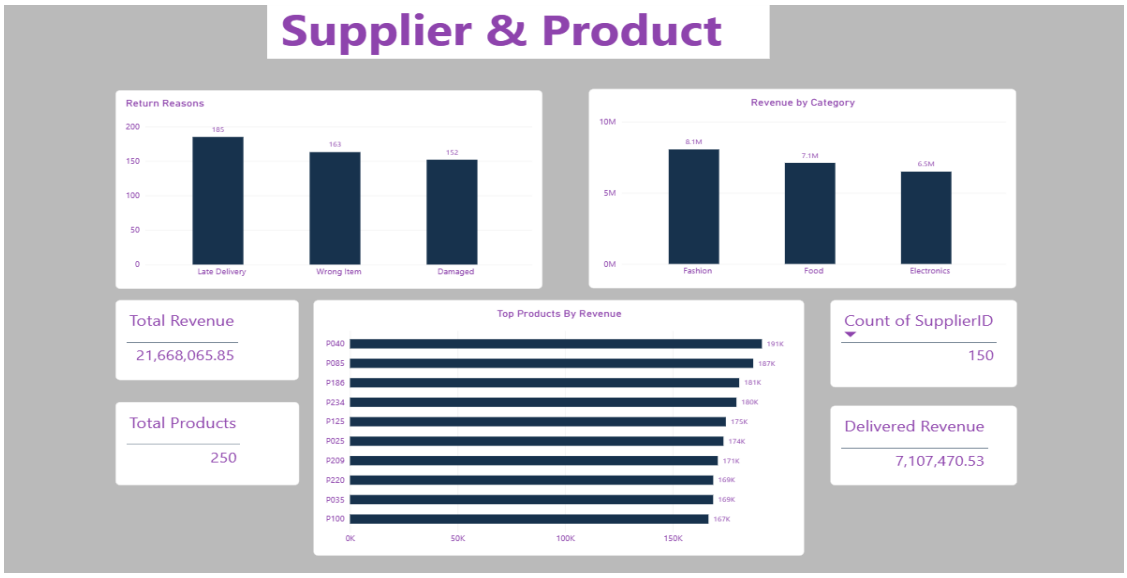
Inventory & Warehouse Dashboard



Shipment & Delivery Dashboard



Supplier & Product Dashboard



PART 5 — Business Analysis & Data Storytelling

5.1 Which products generate the highest revenue?

P040 in Electronics is the highest-revenue product in the supplied analysis, generating NGN 191,463.68 from 404 units across 73 orders.

5.2 Which customers contribute the highest revenue?

CustomerID	Total Revenue	Count of OrderID	City
C0263	74,102.56	9	Lagos
C0815	70,176.55	12	Ibadan
C0512	69,852.45	10	Kano
C0223	69,397.71	8	Kano
C0597	65,424.70	9	Lagos
C0924	64,157.75	9	Lagos
C0581	62,452.67	15	Ibadan
C0300	61,744.97	12	Lagos
C0030	60,862.76	10	Kano
C0489	60,158.92	12	Kano

5.3 Which warehouses operate most efficiently?

The source data does not contain warehouse capacity, so true warehouse utilization cannot be ranked. Delivery on-time rate and shipments per employee are used as operational indicators instead.

WarehouseID	Count of ShipmentID	Average of DelayDays	On-Time Delivery Rate	Count of EmployeeID	Shipments per Employee
W01	332	3.46	41.27%	15	22.13
W10	334	3.49	40.12%	15	22.27
W07	371	3.47	39.62%	17	21.82
W14	340	3.63	38.53%	9	37.78
W04	354	3.52	36.72%	15	23.60
W11	318	3.44	36.48%	11	28.91
W15	352	3.49	36.36%	10	35.20
W08	332	3.67	36.14%	13	25.54
W02	322	3.44	36.02%	12	26.83
W12	337	3.59	35.91%	10	33.70
W13	279	3.61	35.13%	14	19.93
W06	319	3.61	34.17%	14	22.79
W03	344	3.68	32.56%	17	20.24
W09	332	3.68	32.53%	14	23.71
W05	334	3.79	32.04%	14	23.86

5.4 Which suppliers consistently perform well?

This cannot be answered from the supplied operational data because SupplierID is not connected to products, orders, shipments, delivery performance or purchasing records.

5.5 Which products have the highest return rates?

This cannot be answered defensibly at product level because Returns contains OrderID and Reason but no ProductID or ReturnItemID. Joining returns to all order items would incorrectly attribute one return to every item on the order.

5.6 Which regions experience the most delayed deliveries?

City	Total Shipments	Average of DelayDays	On-Time Delivery Rate	Delayed Shipments
Abuja	1205	3.59	35.60%	776
Ibadan	1240	3.58	36.21%	791
Kano	1287	3.64	34.34%	845
Lagos	1268	3.47	38.96%	774
Total	**5000**	**3.57**	**36.28%**	**3186**

Lagos has the largest on-time delivery rate.

Kano has the largest shipment volume.

5.7 Which months record peak business activity?

Month	Total Orders	Total Revenue
January	594	2,582,349.75
February	577	2,509,115.96
March	597	2,598,402.53
April	638	2,742,192.82
May	612	2,716,516.13
June	579	2,545,277.80
July	595	2,535,240.76
August	654	2,850,476.97
September	154	588,493.13

* September is partial through 8 September and should not be compared directly with full months.

5.8 Which KPIs require immediate management attention?

- On-Time Delivery Rate: 36.28%.
- Average Delivery Delay: 3.57 days.
- Return Order Rate: 9.46%.
- Lagos delivery reliability: 35.74% on time.
- Inventory quality/grain issue, because stock totals are not fully interpretable.

5.9 What operational inefficiencies were identified?

- Low delivery reliability across all regions.
- Lagos combines high volume with the weakest on-time rate.
- Late Delivery is the largest return reason.
- Inventory records have repeated warehouse-product combinations without dates or movement identifiers.
- Supplier performance cannot be monitored from the current data model.
- Warehouse capacity is unavailable, preventing true utilization analysis.

5.10 What business opportunities should SwiftLog pursue?

- Improve delivery reliability and reduce late-delivery returns.

- Use high-value and repeat-customer segments for retention programs.
- Protect availability of high-revenue products.
- Use warehouse-level exception monitoring.
- Strengthen supplier and inventory data capture for future optimization.

PART 6 — Business Recommendations

1. Set a delivery-reliability target and review on-time performance weekly.
2. Investigate Lagos warehouse and last-mile processes because it combines the highest shipment volume with the lowest regional on-time rate.
3. Use warehouse-level delay and on-time metrics to target operational interventions.
4. Attack late-delivery root causes because Late Delivery is the largest recorded return reason.
5. Protect high-revenue product availability through stronger stock monitoring.
6. Prioritize Fashion as the largest revenue category while adding cost data before making margin decisions.
7. Segment repeat customers for targeted retention and loyalty initiatives.
8. Add SupplierID links between suppliers and operational purchasing/product flows.
9. Add warehouse capacity fields to enable true utilization analysis.
10. Add inventory snapshot or movement dates to enable inventory turnover.
11. Add ProductID or ReturnItemID to Returns to enable product-level return analysis.
12. Add planned and actual delivery timestamps to strengthen delivery-time measurement.